

Alzikrayat – Architecture Report

PHP/MySQL Photo Gallery Web Application

1. Introduction

Alzikrayat is a web-based photo gallery application developed using PHP and MySQL. The purpose of the system is to allow registered users to create accounts, authenticate themselves, upload photographs, provide titles and descriptions, view photographs, and interact with memories through comments. The application follows an MVC-style structure to separate routing, controllers, models, and views.

2. System Architecture

The application follows a Model-View-Controller architecture. The Router receives HTTP requests and maps them to controller actions. Controllers contain application logic and coordinate requests between models and views. Models communicate with the MySQL database through PDO. Views are responsible for presenting the user interface.

The main flow is: Browser → Router → Controller → Model → Database, followed by Database → Model → Controller → View → Browser.

3. Main Components

Router: Handles GET and POST routes and dispatches matching requests to the appropriate controller method.

Controllers: Handle operations such as registration, login, logout, photo upload, gallery operations, and access control.

Models: Encapsulate database operations. The User model manages user records, while photo and comment models manage gallery data and interactions.

Views: Provide the HTML interface for registration, login, the landing page, gallery, photo upload, and photo-related pages.

4. Authentication and Security

User passwords are stored using PHP password hashing rather than plain text. Login uses password verification and creates a server-side session containing the

authenticated user information. Logout destroys the session. The application also uses CSRF tokens for protected forms and validates user input on the server.

Database operations use PDO prepared statements, reducing the risk of SQL injection. Uploaded files are handled through controlled upload logic and the interface provides HTML5 and JavaScript validation in addition to server-side validation.

5. Database Architecture

The database contains three principal tables: users, photos, and comments. The users table stores account information. Each photo belongs to a user through photos.user_id. Each comment belongs to both a photo and a user.

Foreign keys are implemented using InnoDB. The photo-to-user, comment-to-photo, and comment-to-user relationships use ON DELETE CASCADE. Therefore, dependent records are automatically removed when their parent record is deleted.

6. Gallery and User Interface

The interface uses Bootstrap to provide a responsive layout. The landing page introduces the application and provides navigation based on authentication status. Logged-in users can see their name, access photo upload functionality, and log out. Visitors can access login and registration.

The gallery supports multiple presentation styles, including three-column, four-column, and list layouts. JavaScript changes the presentation without requiring a new server request.

7. Validation

Validation is implemented at multiple levels. HTML5 attributes provide browser-level validation. JavaScript performs client-side checks before form submission. PHP performs server-side validation so that submitted data is checked even when client-side validation is bypassed.

8. Conclusion

Alzikrayat combines PHP, MySQL, MVC-style organization, authentication, file uploads, database relationships, validation, security controls, and a responsive Bootstrap interface in one web application. The architecture separates responsibilities between routing, application logic, data access, and presentation, making the project easier to maintain and extend.